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Chip Floor planning considerations

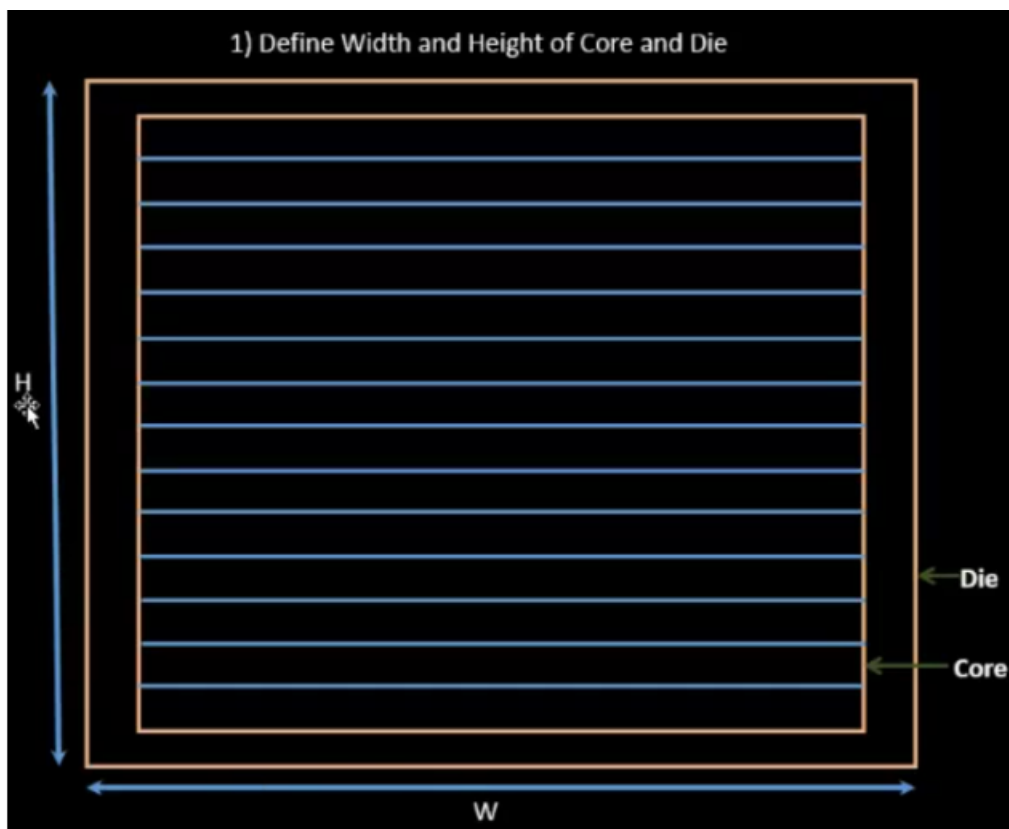
Utilization factor and aspect ratio

This section explains how to determine the width and height of the **core** and **die** in the context of physical integrated circuit (IC) design. It introduces fundamental concepts and theoretical principles required for estimating these dimensions, focusing on the roles of standard cells, netlists, utilization factor, aspect ratio, and the structural relationship between the core and die.

Core Concepts

Physical Design: Core and Die

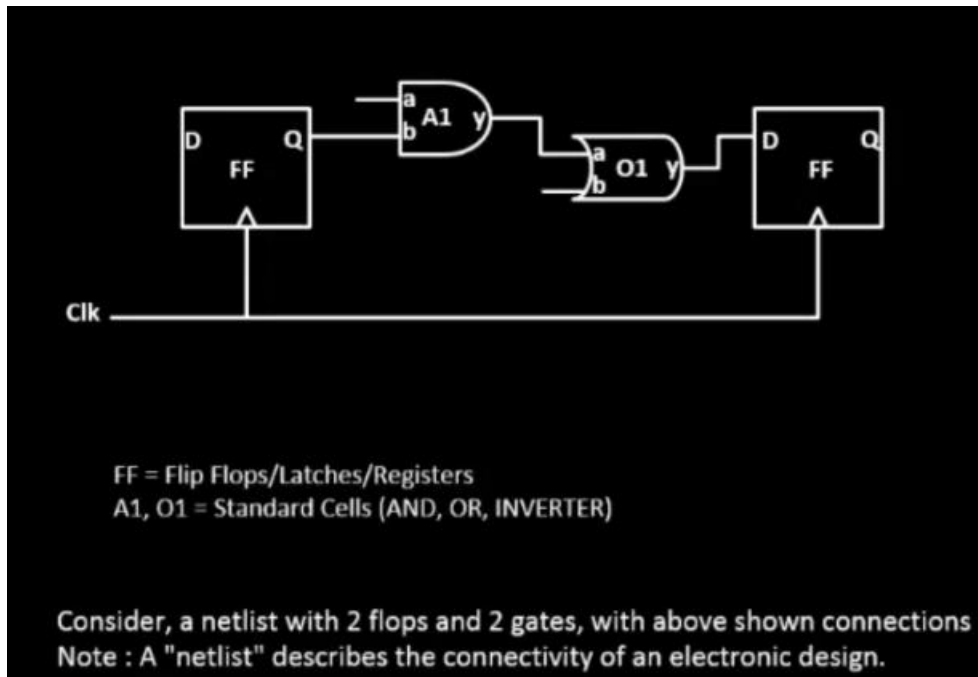
- **Core:** The portion of the chip where fundamental logic and functional circuits (standard cells, registers, memory macros) are placed.
- **Die:** The entire silicon chip area, including the core and additional peripheral regions such as IO rings, power rings, and scribe lines. The die encapsulates the core and provides physical containment for the whole circuit.



Netlist and Standard Cell Dimensions

- The **netlist** specifies the connectivity between components such as flip-flops and logic gates (e.g., AND, OR gates).

- **Standard cells** are pre-designed logic blocks with defined physical dimensions. Their area is typically measured in units based on the manufacturing technology (e.g., 1 unit × 1 unit).
- The **flip-flops** and logic gates are considered standard cells for estimation purposes.

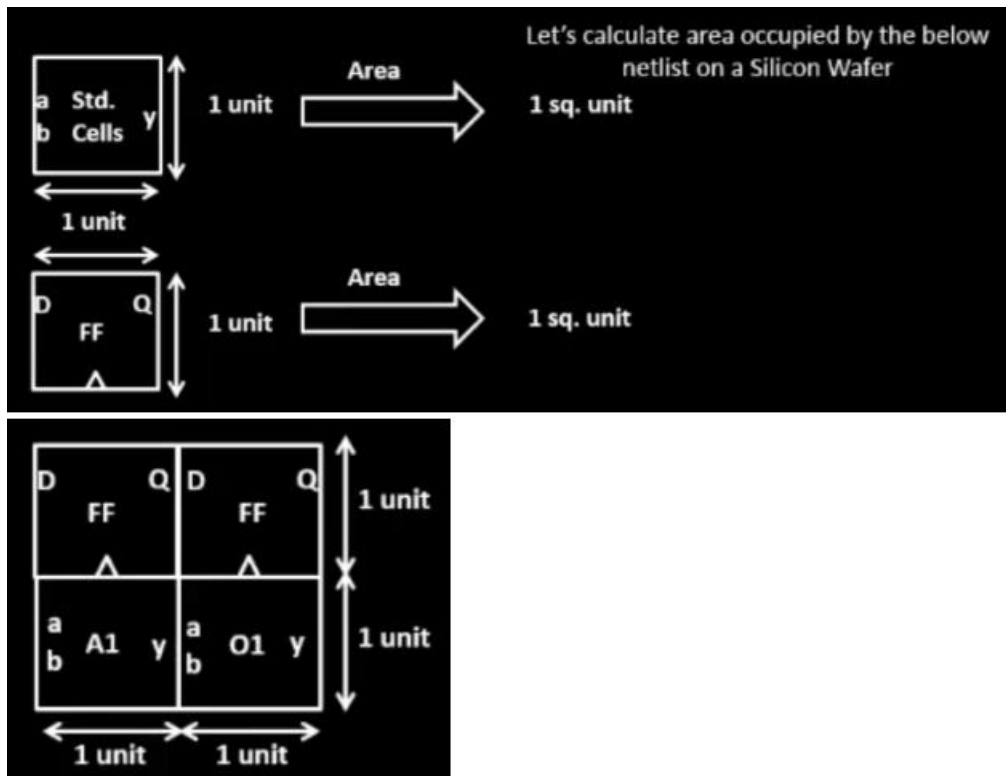


Calculating Area

- The **area occupied by the netlist** is the cumulative area required to place all standard cells (ignoring wires at this stage).
- Placing all standard cells together provides a minimal bounding rectangle (or square) which determines the minimum area required for the design (Area = Length × Width).

Example:

- If four standard cells each of area 1 unit² are arranged, the rectangle would measure 2 units by 2 units, yielding an area of 4 units²



Aspect Ratio

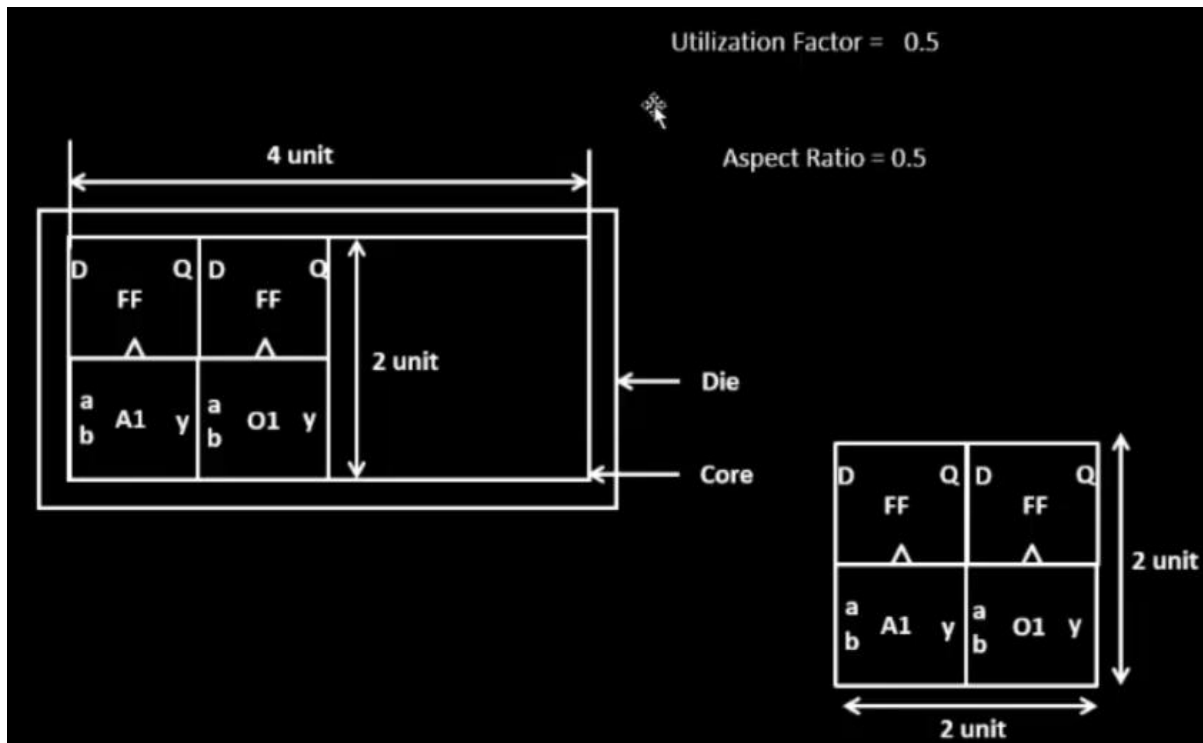
- **Aspect ratio** (AR): Describes the **shape** of the core or die, not its area.
- Defined as:

$$\text{Aspect Ratio} = \frac{\text{Height}}{\text{Width}}$$

- $AR = 1$ implies a square.
- $AR \neq 1$ (e.g., $AR = 0.5$) implies a rectangle.

Example:

- Core of 2 units height & 4 units width: $AR = \frac{2}{4} = 0.5$.



Key Points

- Determining core and die dimensions starts with standard cell area estimates derived from the netlist.
- Netlist area is the sum of standard cell areas, excluding wire routing in early estimation steps.
- Core area must exceed the netlist area, factoring in practical utilization (usually not 100%).
- **Utilization factor** quantifies what fraction of the core is occupied; lower utilization allows for added logic, routing, and buffers during later design stages.
- **Aspect ratio** defines whether the core/die is square or rectangular, impacting floorplanning and packaging efficiency.
- The die includes both the core and additional regions needed for IO and scribe lines, and multiple die are fabricated per wafer.

• Concept of pre-placed cells

- This documentation focuses on the concepts of core area definition, utilization factor, aspect ratio, and the modular approach of preplaced cells in digital hardware, emphasizing theoretical understanding rather than implementation details. Each section explains foundational principles in open-source hardware floor planning and modular circuit design.

- **Core Concepts**

- **Core Area, Utilization Factor, and Aspect Ratio**

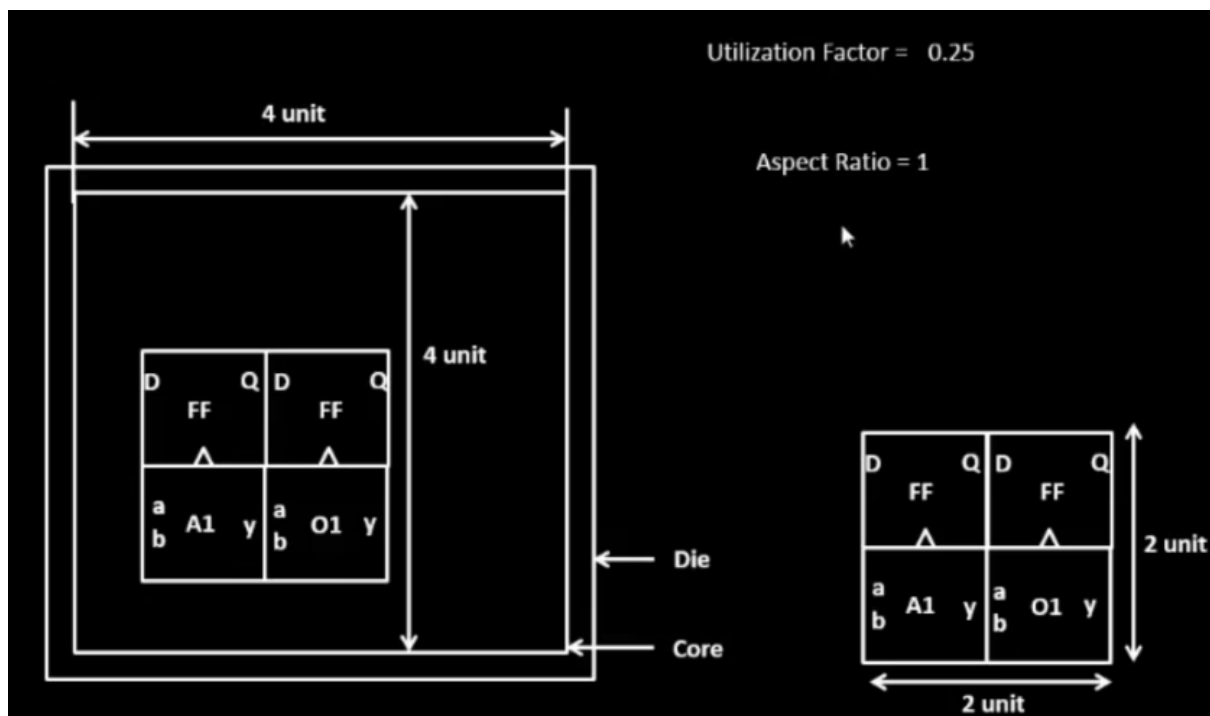
- The core of a chip refers to the central region where the primary functional circuit components are placed. The dimensions of the core are its width (W_{core}) and height (H_{core}).

- **Utilization Factor:**

The utilization factor measures the fraction of the total core area occupied by placed components ("netlist area") versus the total available core area.

- $Utilization\ Factor = \frac{Area\ occupied\ by\ netlist}{Total\ core\ area}$

- For example, if a netlist takes up 4 square units in a $4 \times 4 = 16$ square unit core, the utilization factor is $4/16 = 0.25$, or 25. This means only 25 of the core is used, while the remaining 75 is available, often reserved for routing and potential optimizations.



Aspect Ratio:

The aspect ratio defines the shape of the core and is calculated as:

$$Aspect\ Ratio = \frac{Height}{Width}$$

For a square chip, $Aspect\ Ratio = 1$. Maintaining certain aspect ratios can be critical for performance and manufacturability.

Layering and Routing

Unused area in the core is not wasted. It provides space for routing signal wires between placed components. Routing often occurs across multiple layers above the

placement layer, allowing for more intricate connectivity without increasing the planar area.

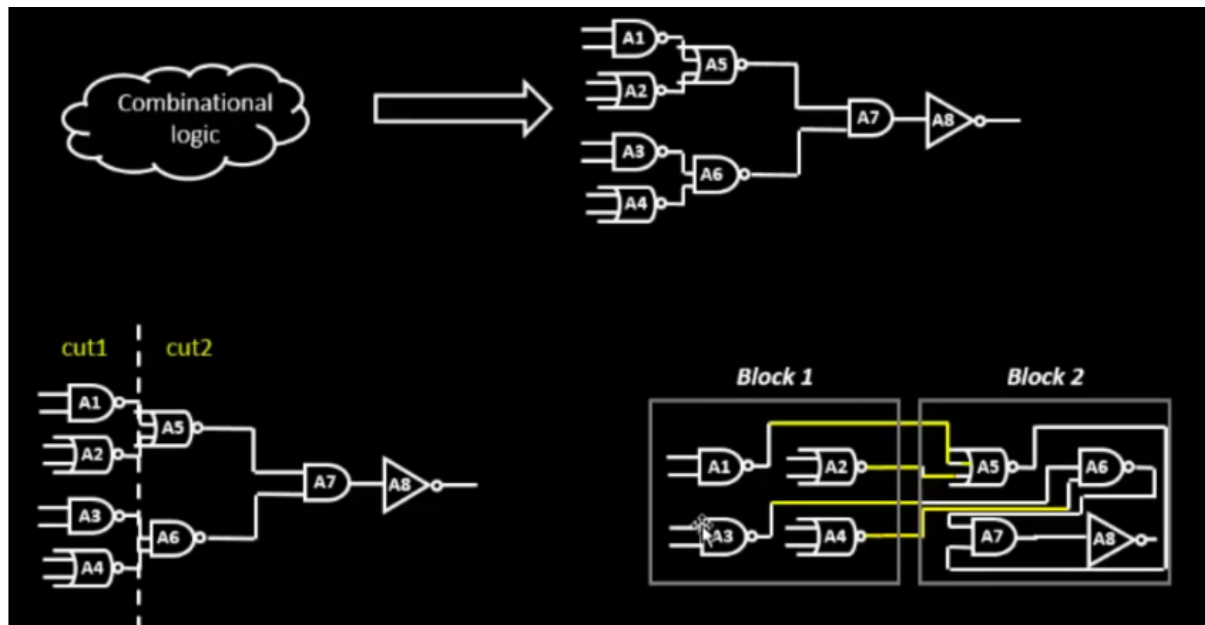
Modular Circuit Design and Preplaced Cells

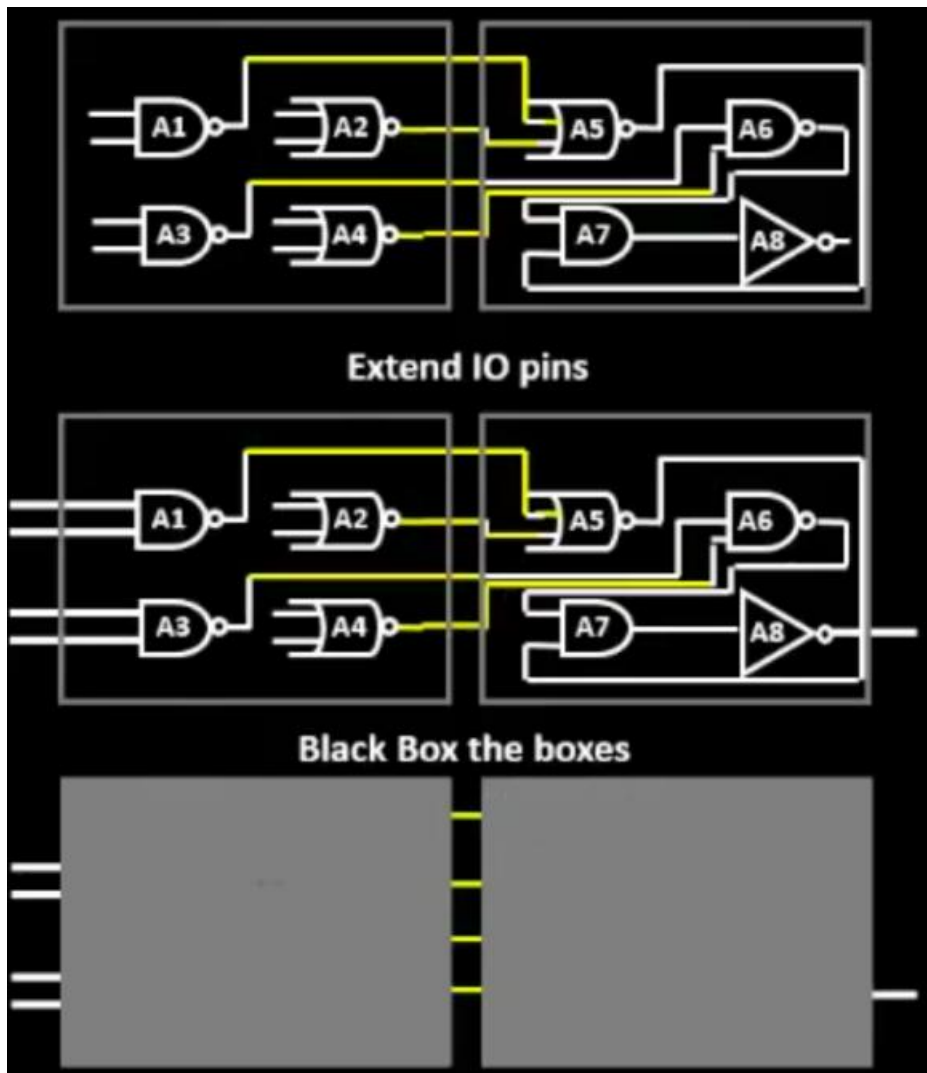
To manage complex designs and increase reusability, large logic blocks are modularized and can be separated from the main circuitry. These modules are often called **preplaced cells**.

Granularization and Blackboxing:

A large combinational block (e.g., 100k gates) may be split into smaller independent modules (e.g., two 50k gate blocks). The internal structure of each module can be "black-boxed"—details are hidden from the top-level view, exposing only input and output (IO) ports.

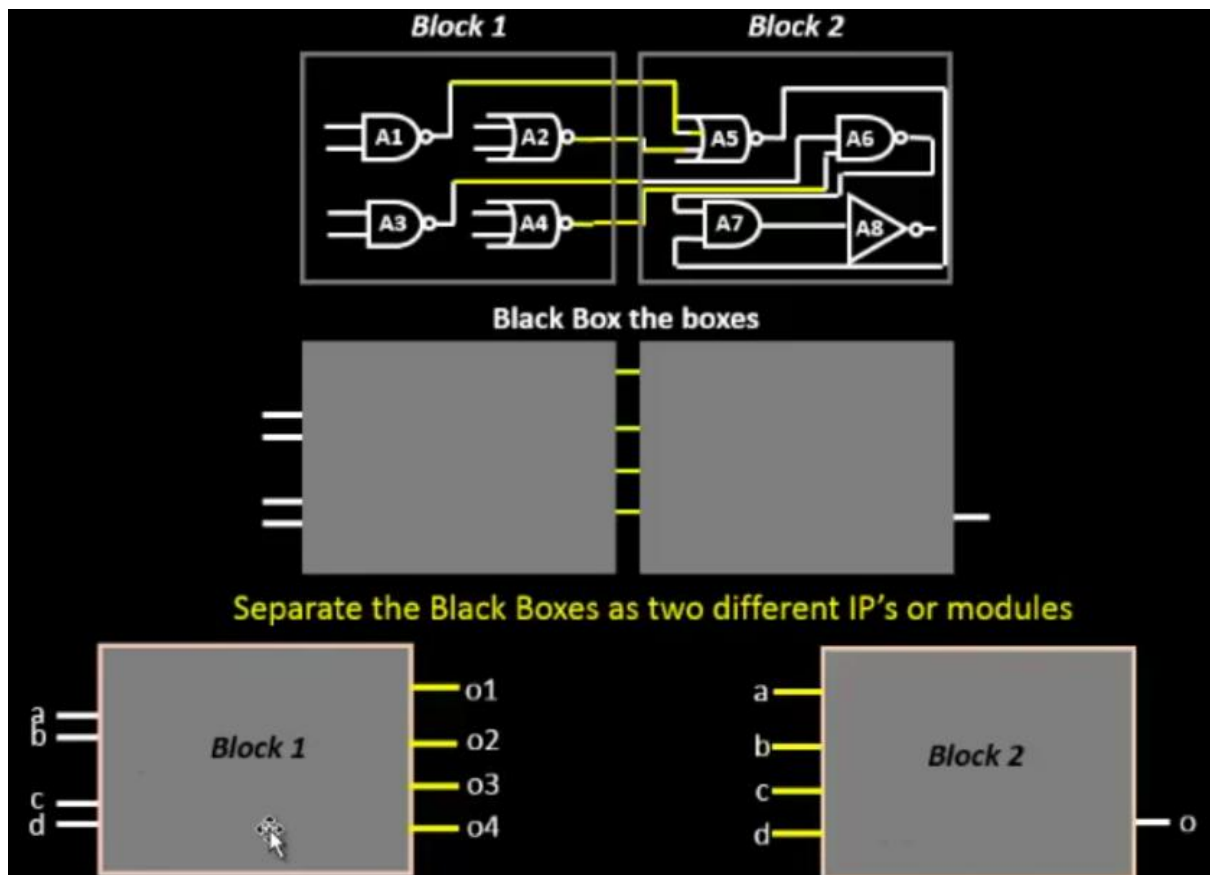
This abstraction means that, from the perspective of the top netlist, the internal logic is invisible, and only the connectivity via IO pins matters.





Reusability:

Once a module is defined and black-boxed, it can be instantiated multiple times within a design. There is no need to duplicate the logic physically; the block is implemented once and reused wherever needed. This is key for components like memories, arithmetic units, or other intellectual property (IP) blocks.



Preplaced Cell Placement:

These cells are physically placed at fixed locations on the chip before automated placement and routing. Their positions remain unchanged throughout further design flows, ensuring predictable timing and connectivity. The placement tools recognize these as fixed, immovable obstacles.

- Similarly, there are other IP's also available, for eg.



- The arrangement of these IP's in a chip is referred as Floorplanning
- These IP's/blocks have user-defined locations, and hence are placed in chip before automated placement-and-routing and are called as pre-placed cells.
- Automated placement and routing tools places the remaining logical cells in the design onto chip

Key Points

- **Utilization factor** quantifies how much of the chip's core area is actively used by logic cells, influencing efficiency and routing capacity.
- **Aspect ratio** of a core impacts the physical layout's effectiveness; a value of 1 indicates a square, which is often optimal for symmetry and ease of design.
- **Unused area** is essential for routing interconnections; multiple routing layers are used for complex wire patterns without expanding the 2D area.
- **Preplaced cells** are large, reusable logic blocks (often IPs) placed at fixed positions pre-layout to ensure modularity and reliability.
- **Blackboxing** abstracts module internal details, exposing only IO pins for top-level interconnection, crucial for managing complexity in large designs.
- **IP reuse** reduces development cost and time by allowing blocks to be implemented once but instantiated multiple times within a circuit.
- **Fixed placement of preplaced cells** guarantees that critical blocks retain their location for optimal physical and timing characteristics, unaffected by automated tools during placement and routing.

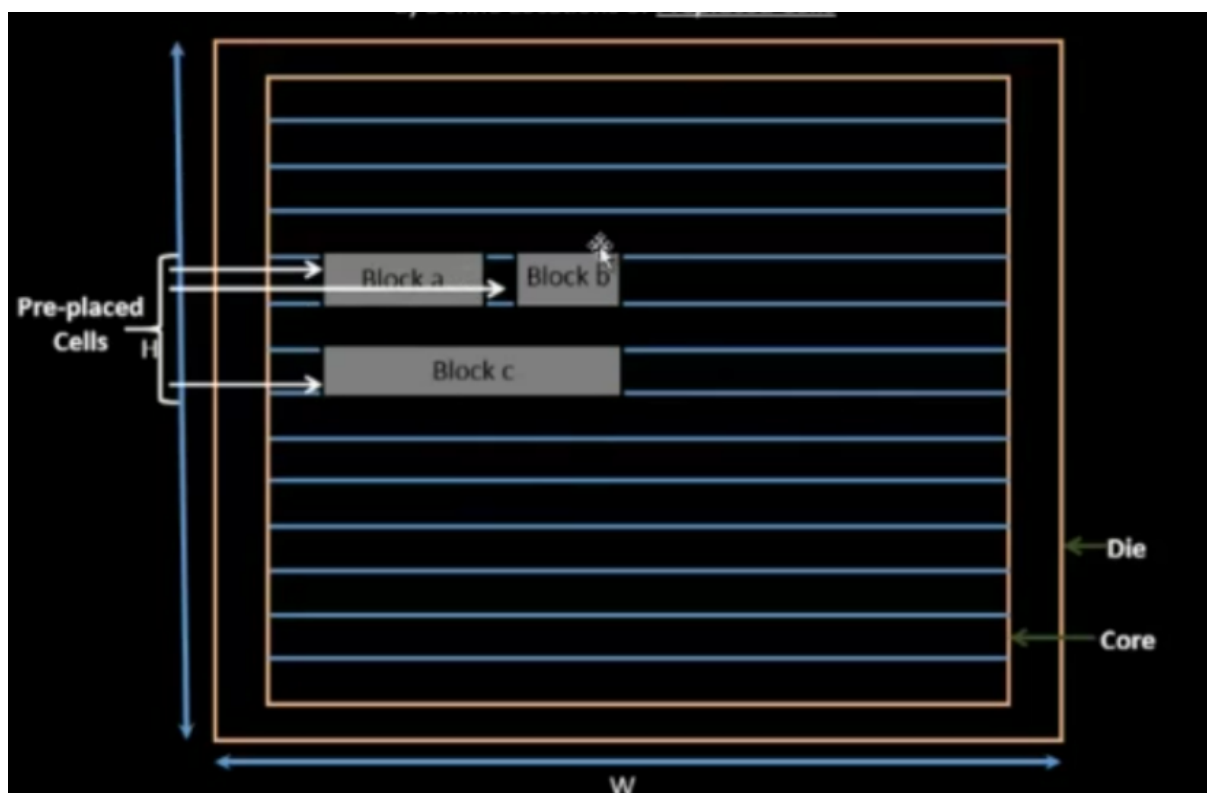
De-coupling capacitors

Preplaced cells are essential building blocks in digital hardware design, implemented as reusable logic modules (e.g., memories, IP blocks) that are instantiated multiple times within a chip. Their locations are fixed prior to automated placement and routing, ensuring optimal connectivity and predictable physical behavior. Effective integration of preplaced cells includes placing them strategically according to the overall design scenario and augmenting them with decoupling capacitors to ensure robust power delivery and noise immunity.

Core Concepts

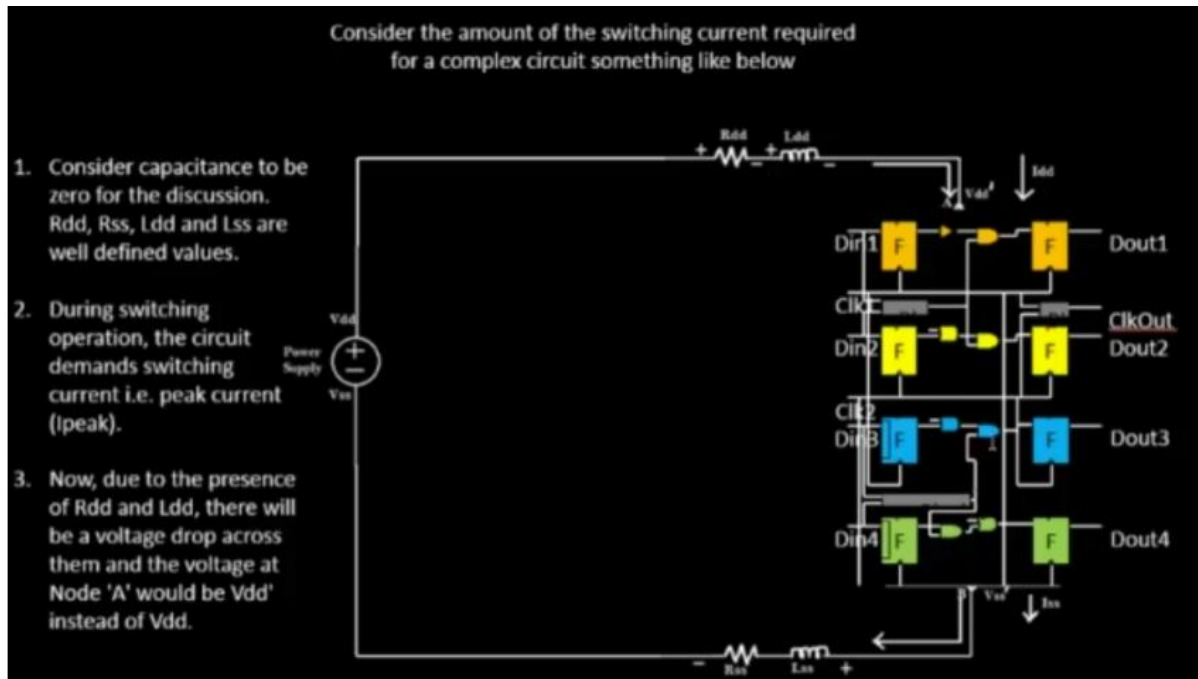
Preplaced Cells (Macros, IPs, Memories)

- **Definition:** Preplaced cells refer to complex logic blocks—such as memories, clocking cells, or IP modules—that are designed once and reused multiple times in a chip.
- **Purpose:** These cells centralize complex functionality and enhance design reuse, reducing redundant effort and enabling modular design strategy.
- **Placement Characteristics:** Their locations are defined before general cell placement and routing. Once set, these positions are fixed and remain untouched by subsequent automated design tools.



Design-Driven Location Assignment

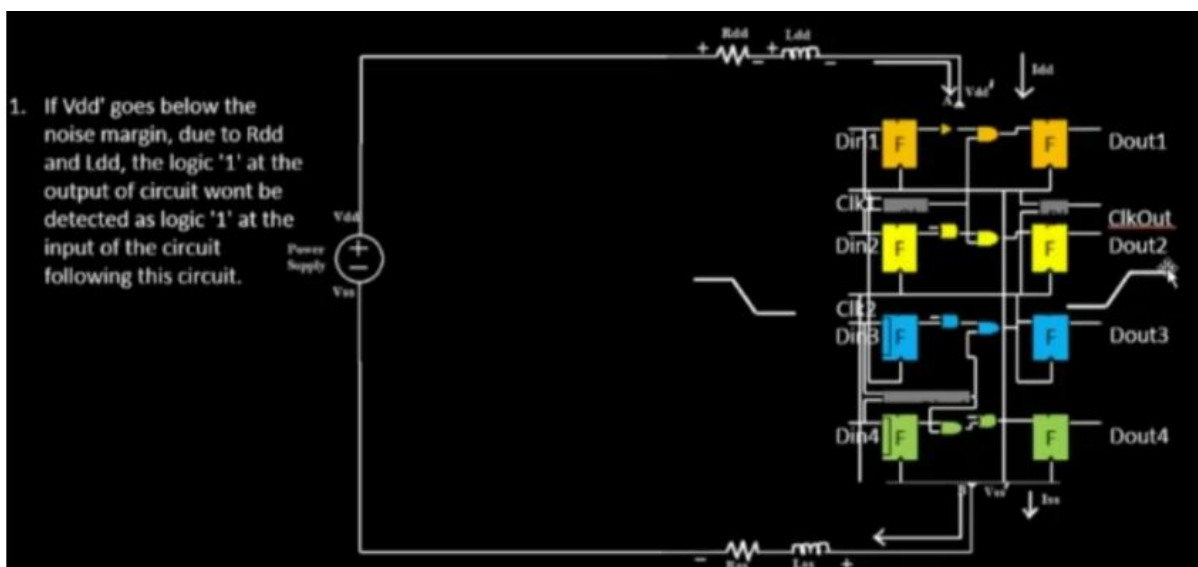
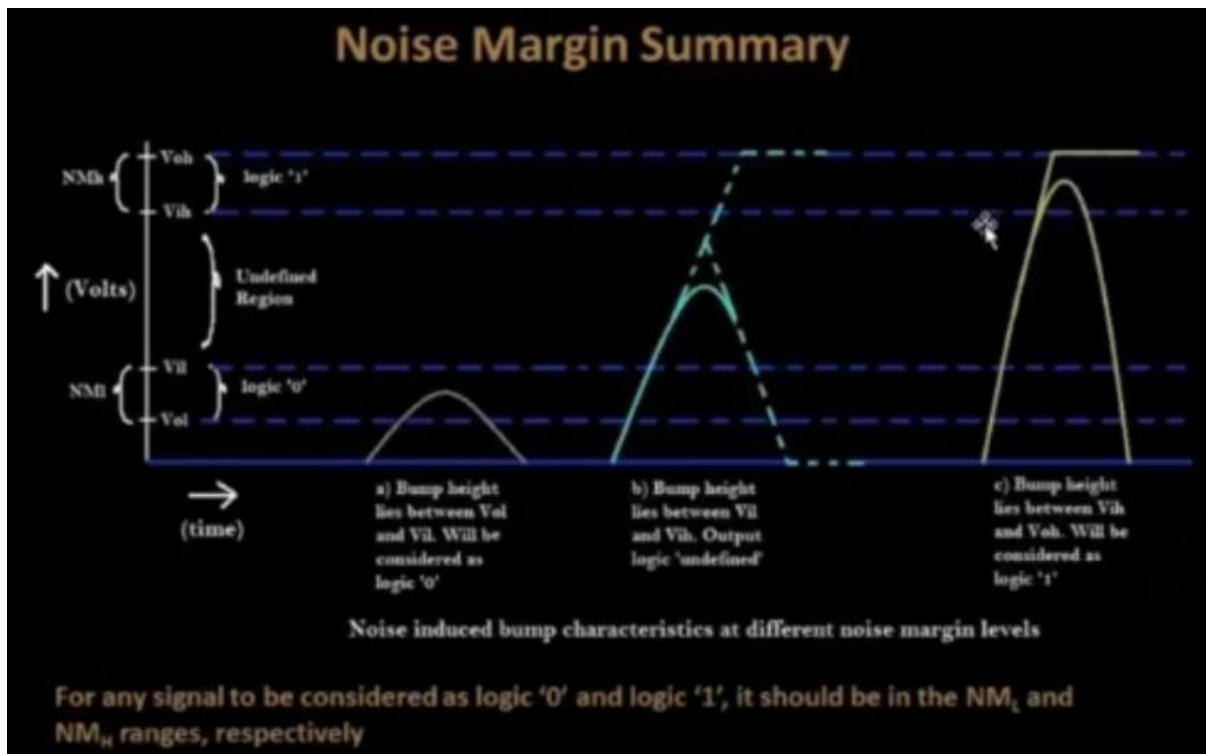
- **Design Scenario:** Placement of preplaced cells aligns with system-level connectivity. For example, blocks interacting heavily with input pins are positioned close to the input edge of the chip to minimize signal latency and congestion.
- **Immutability:** After placement, these block locations do not move throughout the rest of the design flow, ensuring stable timing and predictable power distribution.

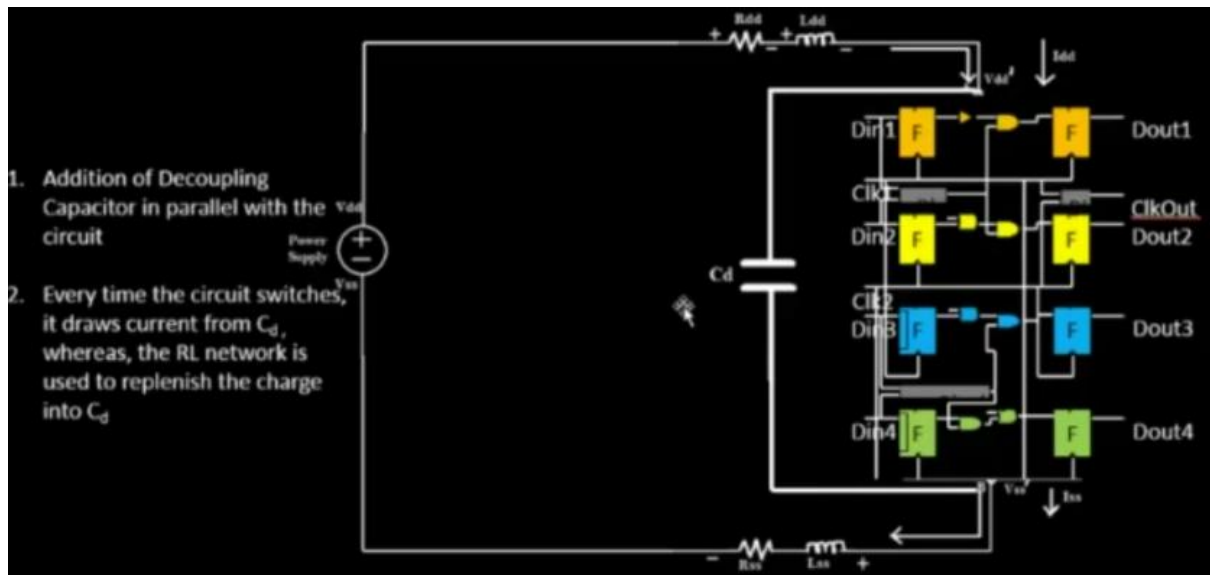


Decoupling Capacitors and Power Integrity

- **Purpose:** Decoupling capacitors are placed around preplaced cells to maintain stable voltage levels during circuit switching events, counteracting voltage drops caused by resistance and inductance in long supply wires.
- **Charging Dynamics:** When a digital circuit switches from logic 0 to logic 1, its internal capacitance demands a surge of current, which is ideally supplied by proximate decoupling capacitors to avoid voltage sag at the logic gate.
- **Voltage Drop and Noise Margin:** The supply voltage delivered over physical wire exhibits a reduction (V_{DD}') due to resistance and inductance. If the voltage at a logic gate drops below the defined noise margin threshold, signal integrity is at risk.
- **Noise Margin Diagram:**
 - Logic '0' is represented by voltages between V_{OL} and V_{IL} .

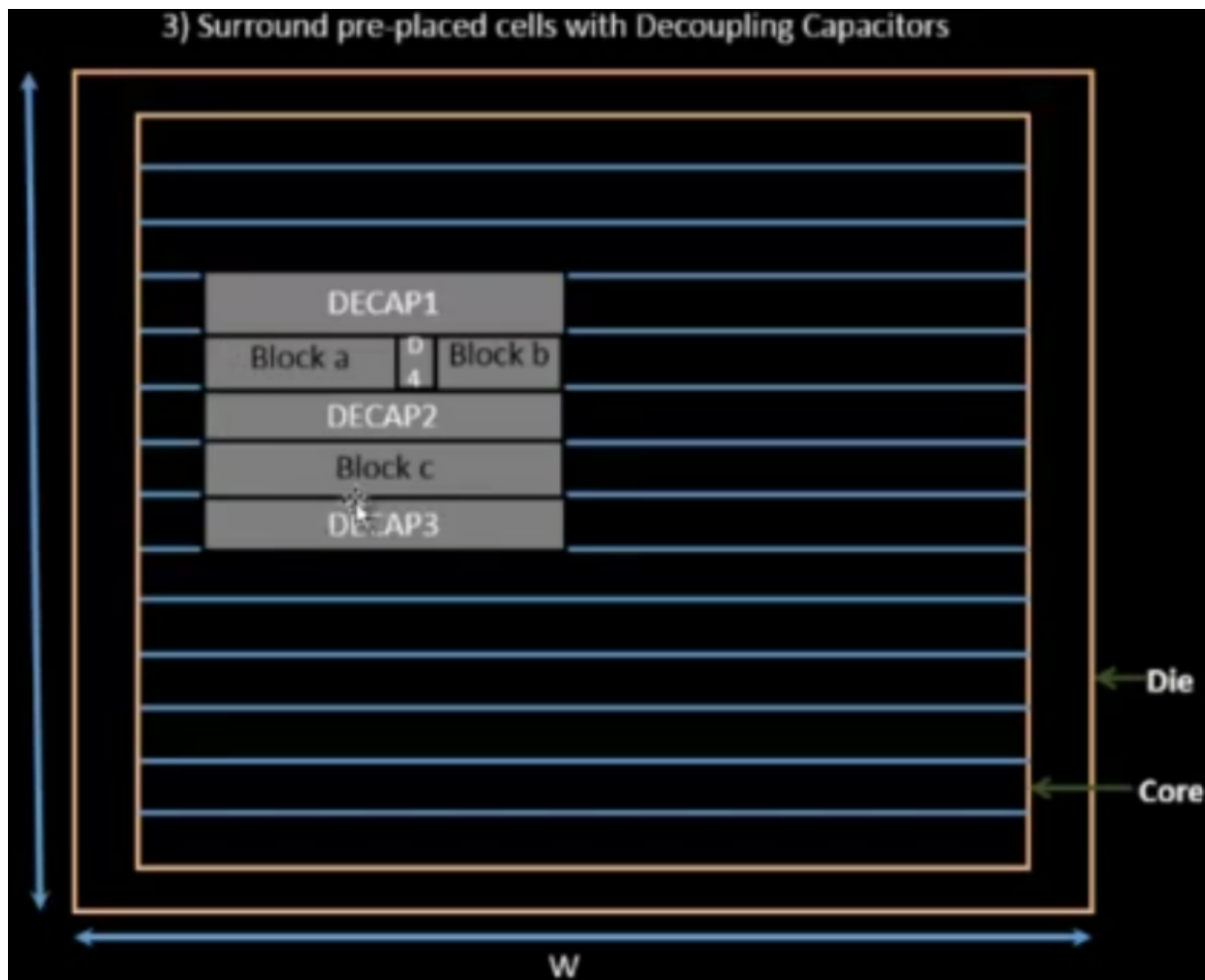
- Logic '1' is voltages between V_{IH} and V_{OH} .
- The region between V_{IL} and V_{IH} is undefined, and signals falling here can result in unpredictable circuit behavior.
- **Supply Assurance:** By placing decoupling capacitors close to preplaced cells, the path resistance and resulting voltage drop are minimized, preserving the full V_{DD} at critical switching moments.





Key Points

- **Placed cells** are complex, reusable logic blocks instantiated with fixed locations, which streamlines design, ensures consistency, and supports large hierarchical designs.
- **Fixed placement:** Once defined, the positions of placed cells are not modified during subsequent automated placement and routing steps.
- **Strategic positioning** is informed by connectivity to primary inputs/outputs and other high-traffic areas, which enhances both performance and routability.
- **Decoupling capacitors** are critical adjuncts, locally supplying charge during switching events and ensuring that core logic operates within defined noise margins.
- The interplay between **voltage drop, supply impedance, local capacitance, and noise margin** underpins the need for carefully engineered power delivery in proximity to placed cells.
- **Global power planning** and grid design (discussed separately) extend these principles to the chip-wide level, ensuring overall power integrity in addition to local delivery.



Power planning

This documentation covers theoretical foundations and key concepts relevant to power planning in modern open-source hardware, focusing on the management of current demands across multiple circuit elements, the limitations of decoupling capacitors, phenomena such as ground bounce and voltage droop, and the use of distributed power meshes as a robust solution for voltage stability in integrated circuits.

Core Concepts

Current Demand and the Role of Decoupling Capacitors

Each electronic element or block within a circuit requires current to function. A **decoupling capacitor** is often incorporated to locally supply charge and mitigate voltage fluctuations caused by these varying current requirements. Decoupling capacitors help smooth out rapid transient demands by storing charge close to the load, ensuring local voltage stability during fast switching events.

Scaling and Power Distribution in Complex Chips

When a given macro or block is repeated multiple times on a chip, the cumulative current demand increases proportionally. Direct decoupling for every element is

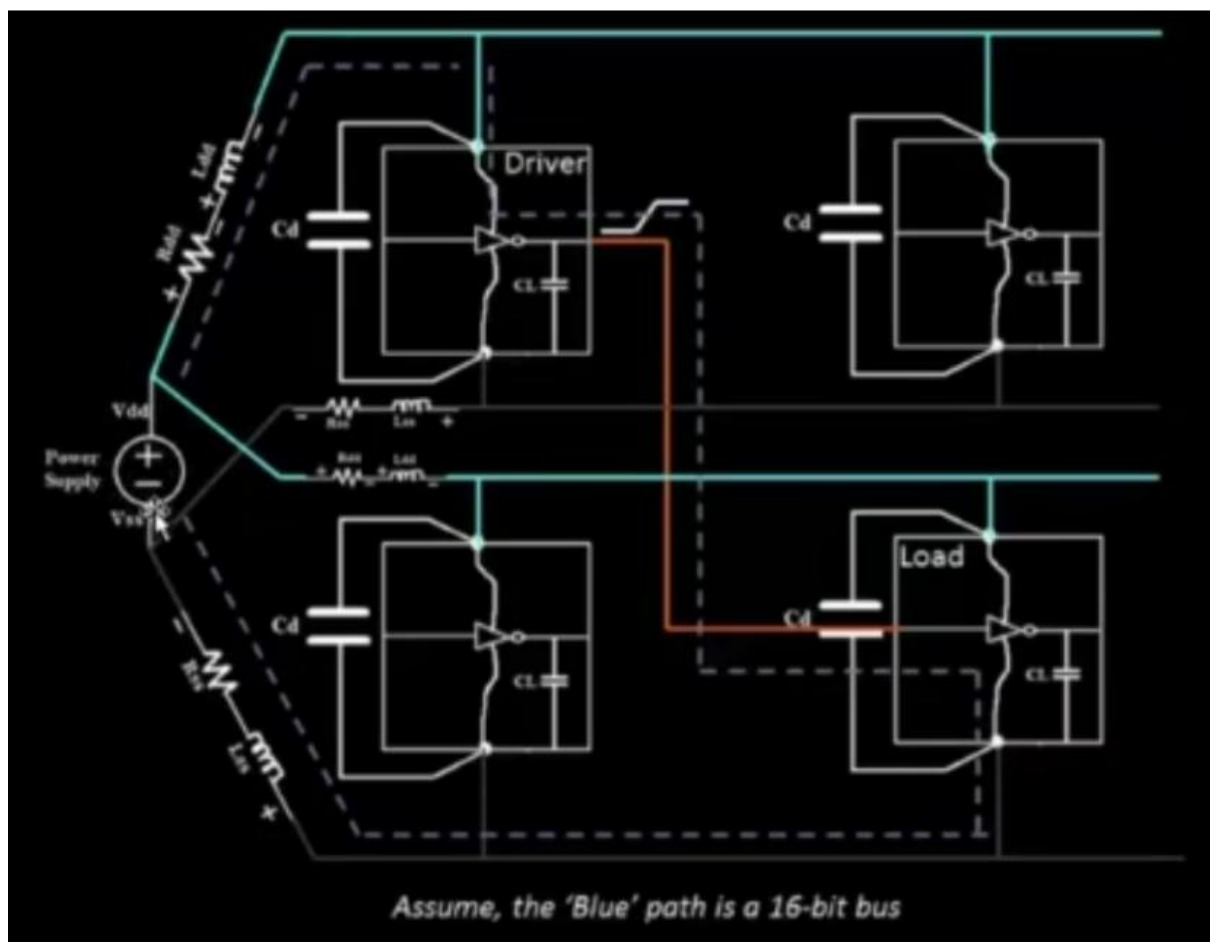
infeasible due to area and cost constraints. Consequently, many elements rely on the central power supply or power distribution network, increasing the importance of effective **power planning** to deliver stable voltage across the entire chip.

Signal Integrity and Voltage Drop

Signal lines (e.g., a 16-bit bus) require clean and consistent voltage levels to maintain signal integrity from driver to load. When the power supply point is physically distant from the signal path, the resistance and inductance of the interconnects may cause a **voltage drop**. As a result, the voltage delivered to the line (V_{actual}) will be less than the nominal supply voltage (V_{DD}):

$$V_{actual} = V_{DD} - I \cdot R_{line}$$

where I is the instantaneous current demand and R_{line} is the resistance of the supply path.



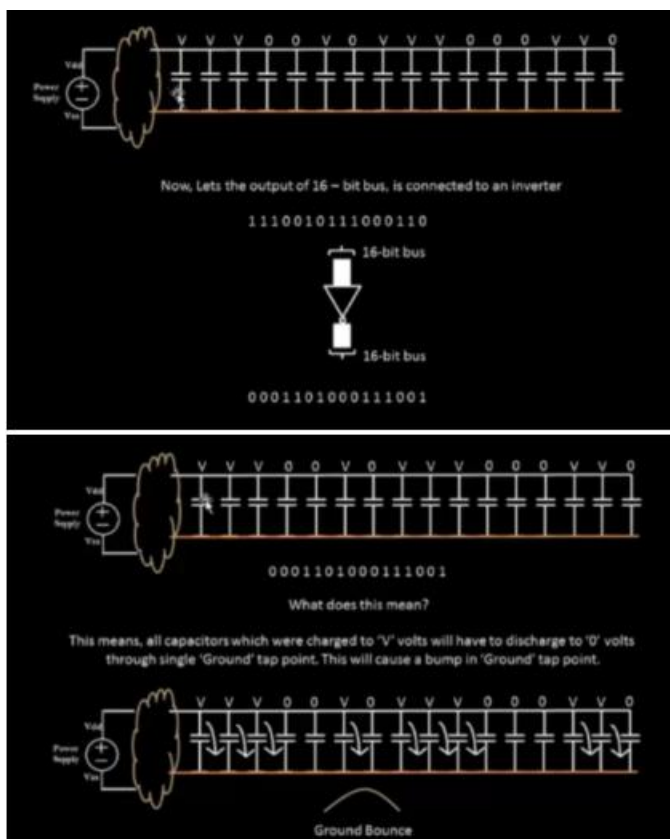
Capacitive Switching and Power Rail Disturbances

Each bit line of a bus can be modeled as a capacitor. When a bus connected to an inverter switches state (all bits transition from 1 to 0 or 0 to 1), all associated capacitors

discharge or charge simultaneously. This simultaneous switching results in large current spikes on either the power or the ground rails.

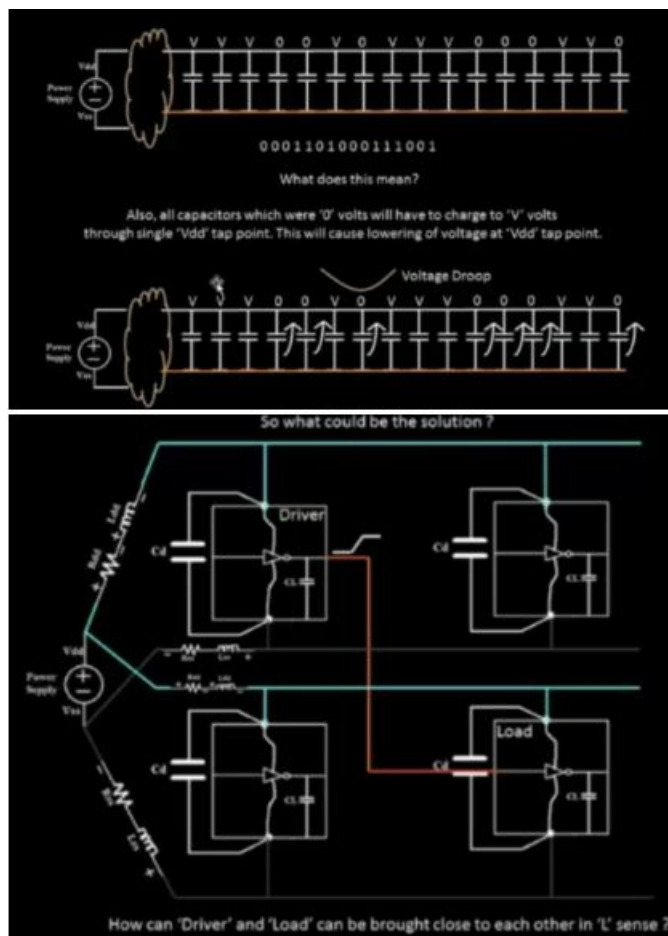
- **Ground Bounce:** When many capacitors discharge to ground at once, the ground potential can temporarily rise above 0V, forming a "bump" or bounce. If this exceeds the designed noise margin, signal levels may enter undefined states.
- **Voltage Droop:** If many capacitors simultaneously charge from the supply rail, the supply voltage may momentarily sag below its nominal value—a phenomenon known as voltage droop.

If these effects exceed allowable thresholds, signal reliability is compromised.



Challenges with Single-Point Power Delivery

Distributing power from a single supply point causes significant voltage gradients and increases the risk of the above problems, particularly as chip size and current demand rise. The single supply must source all the current, resulting in drops and disturbances that propagate through the entire distribution network.



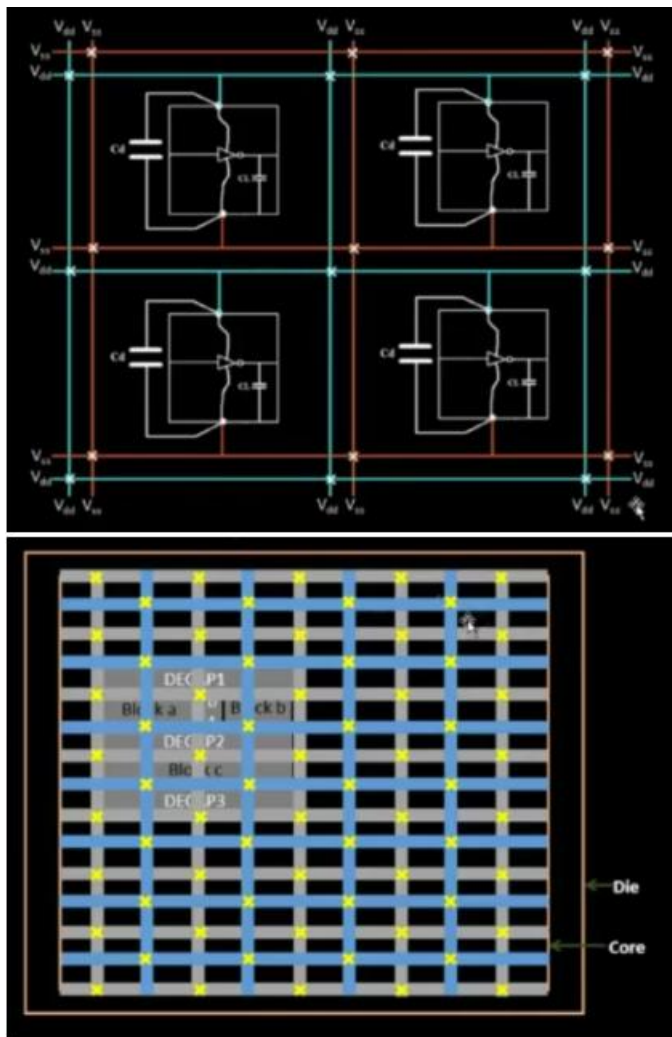
Distributed Power Mesh Networks

The adoption of **power meshes**—interconnected grids of VDD and VSS (ground) lines running horizontally and vertically across the chip—enables each local region to draw current from the nearest supply rather than from a distant source.

Advantages:

- Markedly reduces voltage drop and ground bounce.
- Improves local voltage regulation.
- Enhances the scalability and robustness of large and complex chips.

Any logic block or interconnect can efficiently source or sink current to the nearest point on the mesh, leveraging both vertical and horizontal network structure for minimal path resistance.



Key Points

- **Each circuit element requires a predictable and stable local power supply for reliable operation.**
- **Decoupling capacitors** serve to locally buffer charge but cannot feasibly be deployed for every element in large-scale designs.
- Large, simultaneous switching events (such as on bus lines) can induce voltage anomalies: **ground bounce** and **voltage droop**.
- The magnitude of voltage drops or bounces is proportional to the total current being switched and the impedance of the power path: $V_{drop} = I_{switch} \cdot Z_{path}$.
- If these disturbances exceed the noise margin, digital circuits may operate unpredictably or erroneously.

- **Distributed power mesh networks** (grids of interconnected VDD and VSS lines) fundamentally mitigate these risks by reducing average current path lengths and distributing the load across multiple supply points.
 - Modern VLSI chips utilize such meshes, leading to multiple power pins and entry points distributed throughout the silicon to maintain tight voltage regulation and reduce parasitic effects.
 - Effective power planning and the deployment of power meshes are necessary to ensure both local and global voltage stability, signal integrity, and overall reliability in high-performance, densely packed integrated circuits.
-

Pin placement and logical cell placement blockage

This documentation introduces the **theory and conceptual framework** behind pin placement and connectivity in digital circuit design, using a multi-section flip-flop-based design as a case study. It covers the relationship between netlist connectivity, pin placement strategies, floorplanning, and the critical role of clock signal integrity for high-speed sequential circuits.

![[PLACEHOLDER: diagram/slide/concept-visual]]

Core Concepts

Pin Placement in Circuit Design

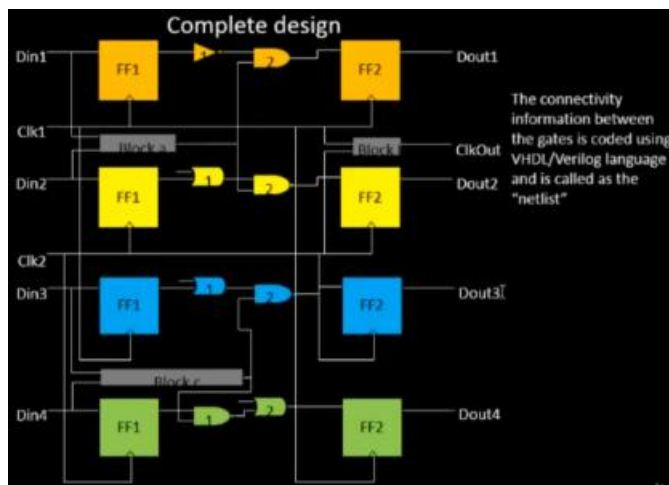
Pin placement is the process of assigning physical locations to input, output, and clock pins on the boundary of an integrated circuit (IC) die. The primary goal is to optimize overall circuit performance, minimize wiring complexity, and ensure robust operation. This is typically handled after the logical circuit design and netlist definition, forming a bridge between the functional (front end) and physical (back end) domains of hardware design.

Netlist and Connectivity

A **netlist** defines the connectivity of logical gates and components within a digital circuit. It specifies:

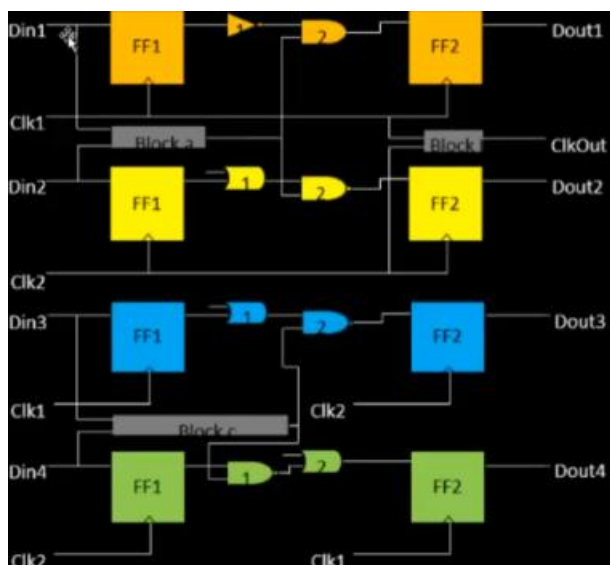
- Which cells, such as flip-flops or logic gates, are present.
- How these cells are interlinked via signals and ports (e.g., clocks, data inputs/outputs).

This connectivity is usually described in hardware description languages (HDLs) like Verilog or VHDL. The netlist is essential for guiding both pin placement and subsequent automated placement and routing steps.



Clock Domains and Inter-Clock Timing

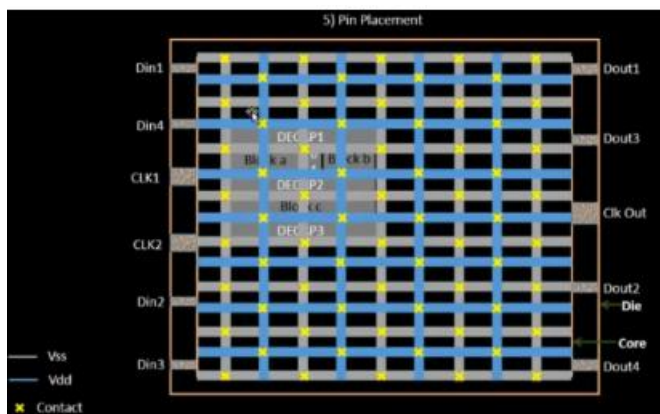
A **clock domain** is a section of a circuit governed by a specific clock signal. Multiple clock domains may exist, driving different flip-flops or logic blocks independently. The scenario where flip-flops are driven by different clocks (e.g., flip-flop 1 by 'clock 1', flip-flop 2 by 'clock 2') is central to **inter-clock timing analysis**. This analysis ensures correct data transfer and synchronization when crossing between domains.



Floorplanning and Pin Arrangement

The **floorplan** is an abstract representation of the chip, detailing block placements and reserved areas for pins:

- Input ports are often placed on the left, output ports on the right, but placements may vary by design needs.
- Physical constraints or block arrangements may dictate pin locations for signal integrity and routing efficiency.
- Port ordering is determined by the proximity to functional blocks, not necessarily by logical sequence.
- Clock pins are physically larger to ensure minimal resistance and high signal integrity because they drive many cells continually.



Pin Resistance and Signal Integrity

In digital circuits, especially those with high fanout or critical timing paths, **clock pins** must have:

- The lowest possible resistance paths to minimize signal degradation.
- Larger physical size to reduce electrical resistance, ensuring robust and reliable clock distribution throughout the chip.

This principle supports high speed, low-latency signal propagation, which is vital for synchronous operations.

$$R = \frac{\rho L}{A}$$

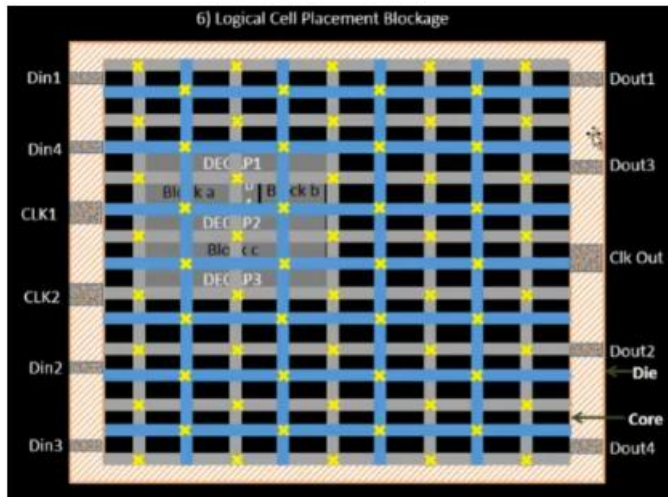
Where:

- R is resistance,
- ρ is resistivity,
- L is wire length,
- A is cross-sectional area.

Increasing A (by making clock pins larger) decreases R .

Logical Blockage and Tool Guidance

After pin positions are finalized, **logical placement blockages** are defined to prevent automated tools from inadvertently placing cells in reserved pin regions. This ensures pins remain unobstructed for connectivity and signal integrity, streamlining the subsequent physical design steps.



Key Points

- **Pin placement optimizes physical connectivity**, minimizes routing complexities, and enhances performance.
- The **netlist** forms the basis for mapping logical connectivity to physical locations.
- **Clock domain interactions** (inter-clock timing) are a major source of timing challenges in multi-clock designs.
- **Floorplanning places ports** (inputs/outputs) according to proximity to relevant functional blocks and routing efficiency, not just logical sequence.
- **Clock pins use larger pads** to lower electrical resistance and ensure robust clock signal distribution.
- **Placement blockages** are defined to reserve areas around pins, guiding automated place-and-route tools.
- **Front-end and back-end teams** must coordinate—the front end defines netlist connectivity, the back end executes pin placement and physical constraints.

Steps to run floorplan using OpenLANE

This lab covers the floor plan stage of the physical design flow in open-source hardware design. The floor plan stage follows synthesis and involves setting up the chip layout by defining the die area, core area, aspect ratio, utilization factor, and placement of input/output cells. Additionally, this stage includes Power Distribution Network (PDN) creation and macro placement. The lab will guide you through configuring floor plan parameters, understanding the configuration hierarchy, and executing the floor plan flow using open-source tools.

Aim: Learn to configure and execute the floor plan stage of the design flow, including understanding switches, configuration files, and parameter settings.

Key Steps:

- Review synthesis flow completion
- Understand floor plan configuration parameters and switches
- Examine design-specific and system-default configuration files
- Set core utilization, aspect ratio, and I/O metal layer parameters
- Execute the floor plan flow and verify successful completion